

Week 7: Network Psychometrics

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1 Introduction

i Learning objectives

By the end of this practical, you should be able to:

1. Explain the core concepts of network modelling, including nodes, edges, network structure, and common network types used in social and behavioural research.
2. Interpret key network metrics (e.g., degree, strength) and describe what they reveal about the structure and dynamics of a system.
3. Construct and visualise basic networks using R.
4. Apply introductory network-analytic techniques to real data, evaluate emerging patterns, and communicate findings using appropriate visual and statistical summaries.

In the lecture, we saw how network models represent psychological variables as nodes, and their conditional dependencies as edges. We also introduced regularisation to estimate sparse, interpretable networks, and bootstrapping to evaluate the stability of estimated edges.

Today, we will apply these ideas to two short questionnaires from a study of parents of children with special educational needs during the COVID-19 pandemic:

- **Part A:** Six items from the Kessler Psychological Distress Scale
- **Part B:** Six items from the Warwick–Edinburgh Mental Wellbeing Scale
- **Part C:** Combining the 12 items and figuring out the overall structure of Mental Health

2 Setting up

2.1 Packages

Install and load the following packages:

```
library(haven)      # import Stata data
library(tidyverse)  # data wrangling
library(psych)      # polychoric correlations
library(bootnet)    # network estimation + bootstrapping
library(qgraph)     # network visualisation
```

```
library(igraph)      # graph tools
library(NetworkComparisonTest) #network comparison test
```

3 Part A: Core Example

3.1 Introduction to the dataset

We begin with six distress items, these are taken from the Kessler Psychological Distress Scale

- nervous
- hopeless
- restless
- nothing_cheer
- everything_effort
- worthless

These are ordinal (1–5). Clear your workspace, set your working directory, and load the data - note that you'll need to use the `read_dta()` function to import this file.

Next, extract the distress items from the dataset:

```
distress <- c("nervous", "hopeless", "restless",
              "nothing_cheer", "everything_effort", "worthless")

items <- data[distress]
nrow(items)
summary(items)
colMeans(items, na.rm = TRUE)
```

1. **How many observations are in the dataset?**
Use `nrow(items)`.
2. **How many missing values does each item have?**
Use `summary(items)`.
3. **Which item has the highest mean?**
Use `colMeans(items, na.rm = TRUE)`.

3.2 Preparing the data

Convert labelled variables to numeric to be sure:

```
items <- data.frame(lapply(items, function(x) as.numeric(as.character(x))))
str(items)
```

```
'data.frame': 517 obs. of 6 variables:
 $ nervous      : num NA 3 2 3 3 4 NA NA 5 NA ...
 $ hopeless     : num NA 3 2 3 5 4 NA NA 2 NA ...
 $ restless     : num NA 4 2 3 4 5 NA NA 3 NA ...
 $ nothing_cheer : num NA 2 5 4 5 5 NA NA 1 NA ...
 $ everything_effort : num NA 2 4 3 4 4 NA NA 1 NA ...
 $ worthless    : num NA 4 5 5 5 5 NA NA 2 NA ...
```

```
summary(items)
```

nervous	hopeless	restless	nothing_cheer
Min. :1.000	Min. :1.000	Min. :1.000	Min. :1.000
1st Qu.:2.000	1st Qu.:3.000	1st Qu.:2.000	1st Qu.:3.000
Median :3.000	Median :4.000	Median :3.000	Median :4.000
Mean :3.072	Mean :3.551	Mean :3.251	Mean :4.105
3rd Qu.:4.000	3rd Qu.:5.000	3rd Qu.:4.000	3rd Qu.:5.000
Max. :5.000	Max. :5.000	Max. :5.000	Max. :5.000
NA's :211	NA's :212	NA's :214	NA's :213

everything_effort	worthless
Min. :1.000	Min. :1.000
1st Qu.:3.000	1st Qu.:3.000
Median :3.000	Median :4.000
Mean :3.326	Mean :4.057
3rd Qu.:4.000	3rd Qu.:5.000
Max. :5.000	Max. :5.000
NA's :216	NA's :221

1. Are all variables now numeric?
2. Have the descriptive statistics changed?

3.3 Polychoric correlations

Compute polychoric correlations:

```
poly_corr <- polychoric(items)$rho
poly_corr
```

	nervous	hopeless	restless	nothing_cheer	everything_effort
nervous	1.0000000	0.4757092	0.4773222	0.3630860	0.2526908
hopeless	0.4757092	1.0000000	0.4255863	0.6428647	0.5692128
restless	0.4773222	0.4255863	1.0000000	0.3832485	0.2842738
nothing_cheer	0.3630860	0.6428647	0.3832485	1.0000000	0.6790467
everything_effort	0.2526908	0.5692128	0.2842738	0.6790467	1.0000000
worthless	0.3996348	0.6874185	0.3870143	0.7040625	0.6546725

worthless

nervous	0.3996348
hopeless	0.6874185
restless	0.3870143
nothing_cheer	0.7040625
everything_effort	0.6546725
worthless	1.0000000

1. Which pair shows the strongest correlation?
2. Which pair shows the weakest?
3. Do these patterns make psychological sense?

3.4 Estimating a partial correlation network

```
net <- estimateNetwork(
  items,
  default = "EBICglasso",
  corMethod = "cor_auto"
)
```

Estimating Network. Using package::function:

- qgraph::EBICglasso for EBIC model selection
- using glasso::glasso
- qgraph::cor_auto for correlation computation
- using lavaan::lavCor

Variables detected as ordinal: nervous; hopeless; restless; nothing_cheer; everything_effort; wor

Warning in EBICglassoCore(S = S, n = n, gamma = gamma, penalize.diagonal = penalize.diagonal, : A dense regularized network was selected (lambda < 0.1 * lambda.max). Recent work indicates a possible drop in specificity. Interpret the presence of the smallest edges with care. Setting threshold = TRUE will enforce higher specificity, at the cost of sensitivity.

1. Why is cor_auto appropriate?
2. What does the sparsity warning imply?
3. What is EBICglasso, and why is it used to estimate the network?

3.5 Inspecting the partial correlation matrix

```
partial_cor_matrix <- net$graph
partial_cor_matrix
```

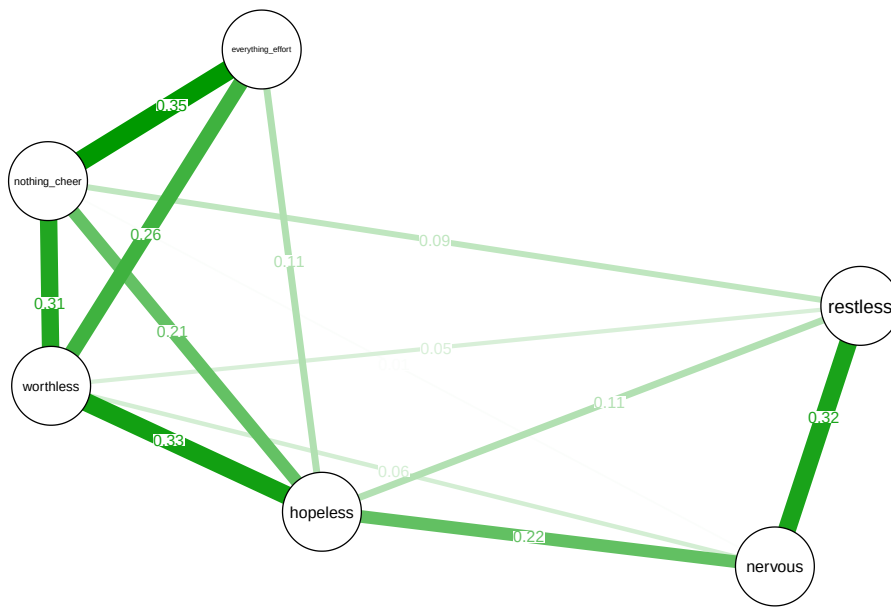
	nervous	hopeless	restless	nothing_cheer
nervous	0.000000000	0.2156853	0.31556873	0.005394736
hopeless	0.215685300	0.0000000	0.10652941	0.214032491
restless	0.315568732	0.1065294	0.00000000	0.087621946
nothing_cheer	0.005394736	0.2140325	0.08762195	0.000000000
everything_effort	0.000000000	0.1077785	0.00000000	0.350987468
worthless	0.060110434	0.3260112	0.05424997	0.305585527

	everything_effort	worthless
nervous	0.0000000	0.06011043
hopeless	0.1077785	0.32601116
restless	0.0000000	0.05424997
nothing_cheer	0.3509875	0.30558553
everything_effort	0.0000000	0.26408804
worthless	0.2640880	0.00000000

1. Which edges are non-zero?
2. Which strong marginal correlations disappear?
3. What does this tell you?

3.6 Plotting the network

```
qgraph(
  net$graph,
  labels      = colnames(items),
  layout      = "spring",
  edge.labels = TRUE,
  edge.label.cex = 0.8
)
```



1. Which symptoms are most strongly connected?
2. Are there clusters?
3. Does the visual structure match the matrix?

3.7 Bootstrapping edge accuracy

```
boot_edges <- bootnet(
  net,
  nBoots = 100,
  type = "nonparametric"
)
```

Note: bootnet will store only the following statistics: edge, strength, outStrength, inStrength

Bootstrapping...

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Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated

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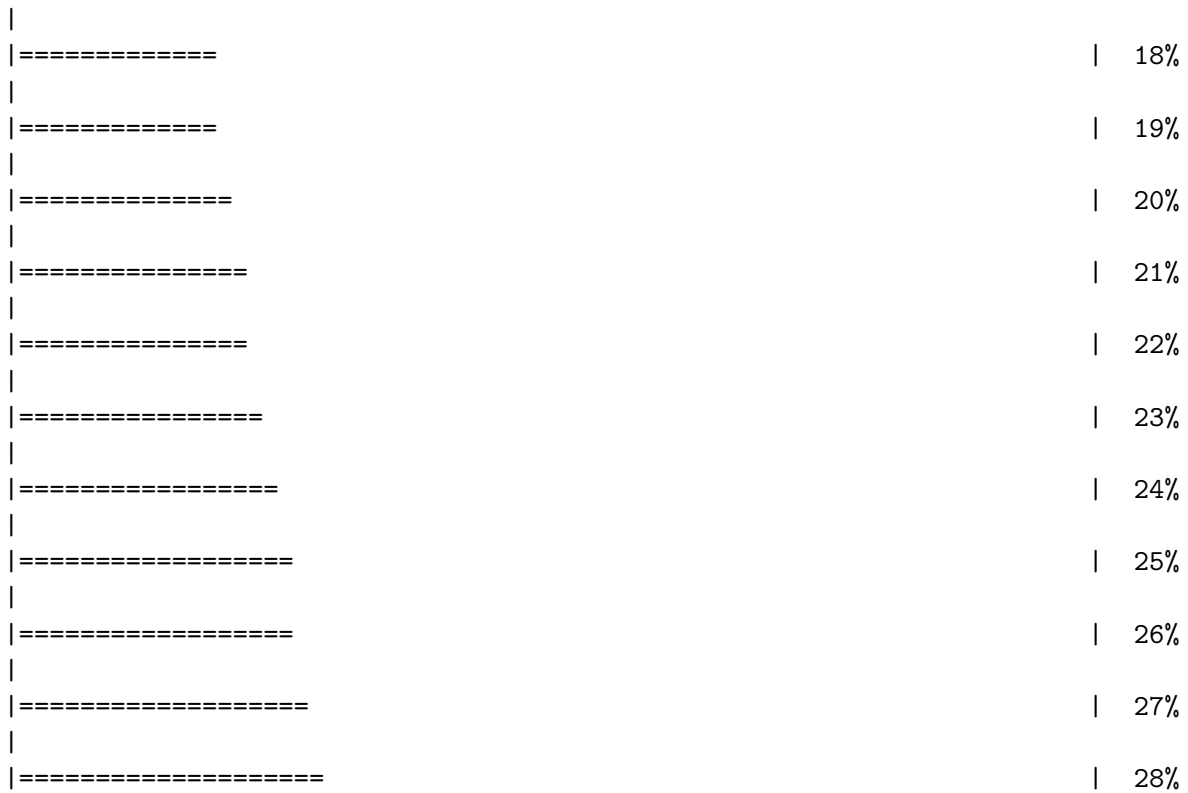
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Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated



Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated



Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated



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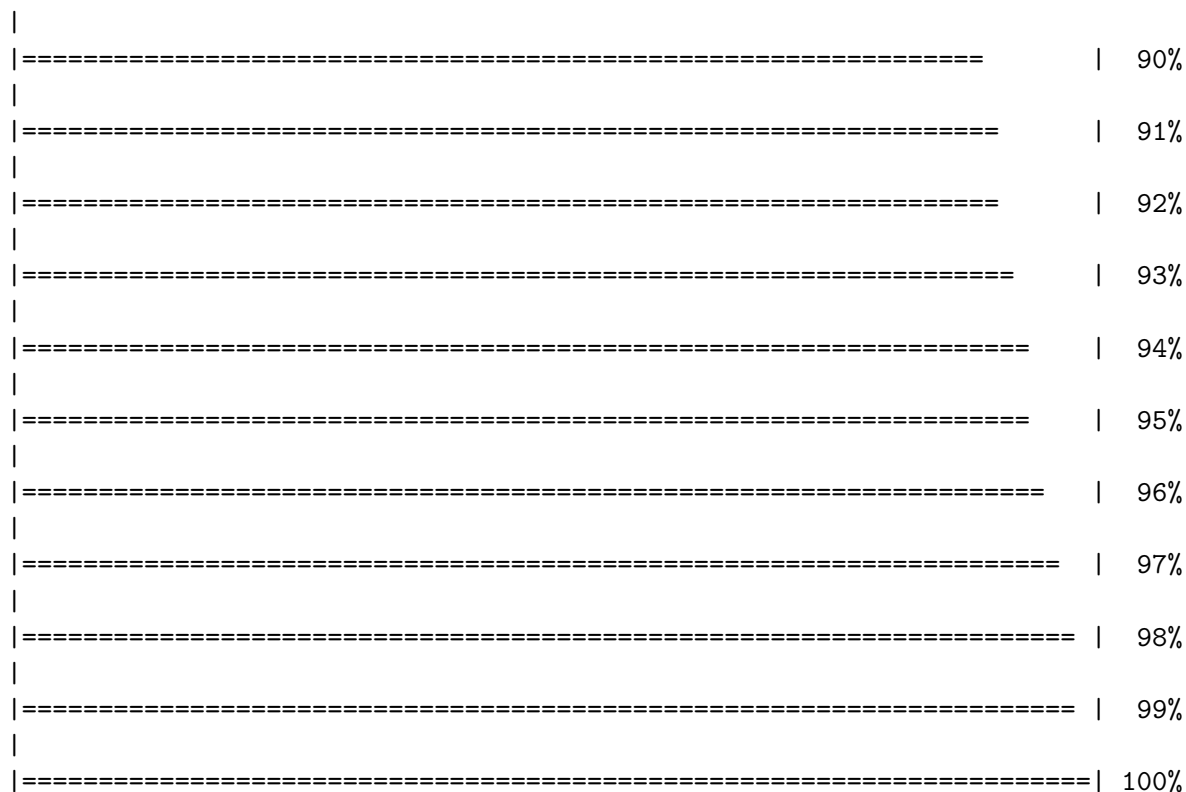
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Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated



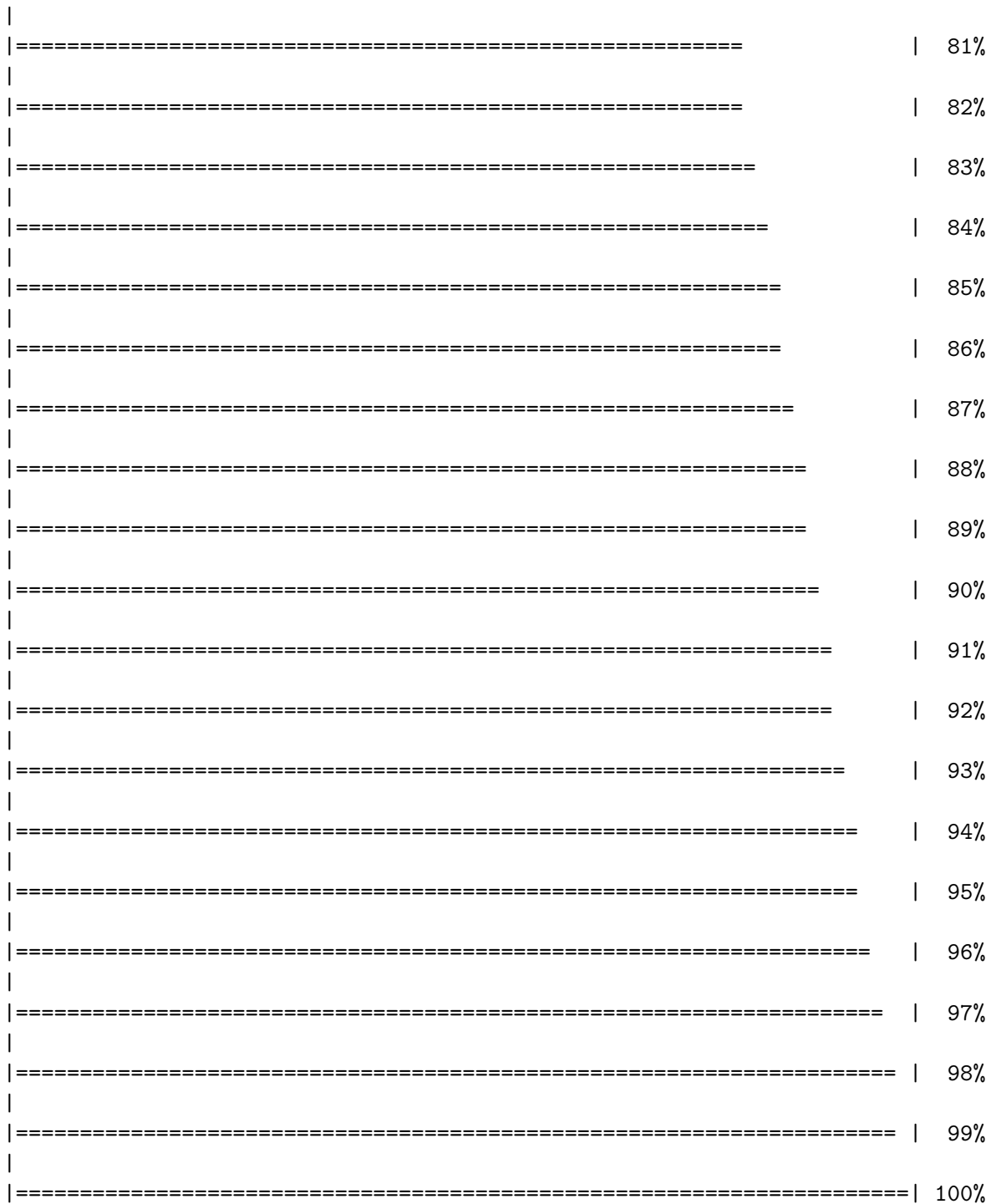
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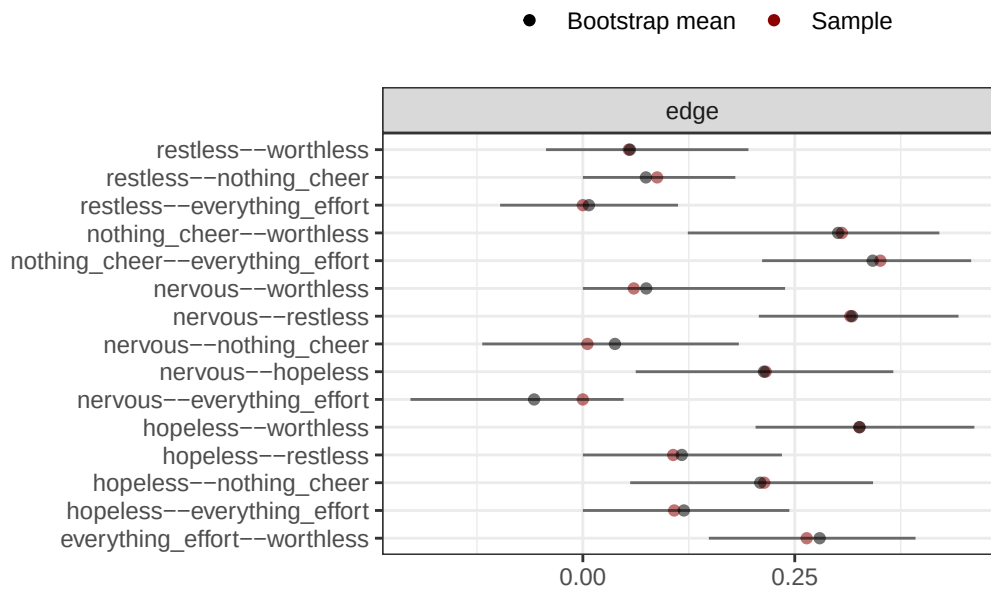
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```
plot(boot_edges, "edge", plot = "interval")
```

1. Which edges have narrow CIs?
2. Which are unstable?
3. How does this affect interpretation?

3.8 Node centrality - Degree & Centrality

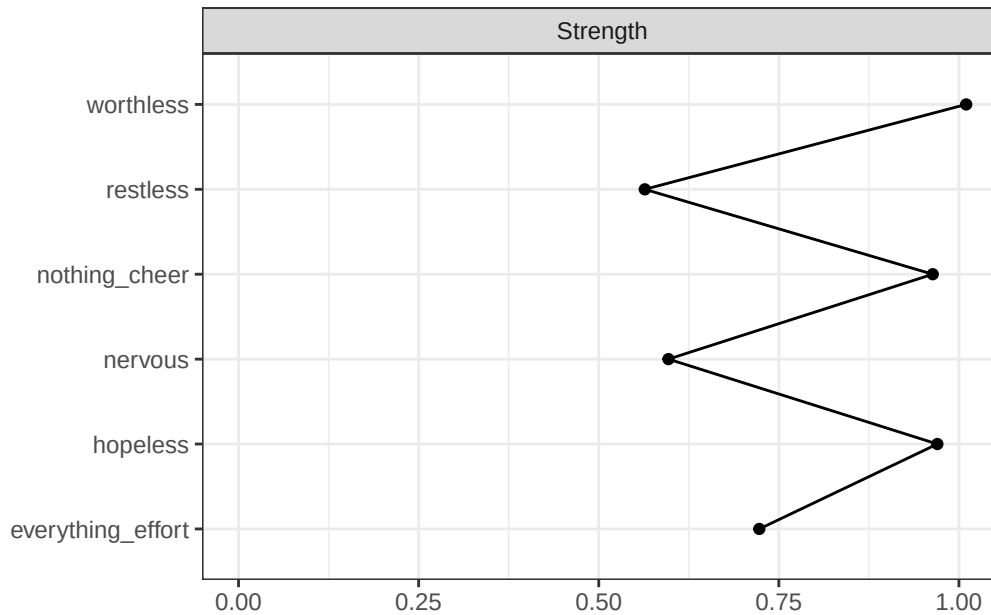
```
# Node degree = number of non-zero edges per node
degree <- apply(net$graph != 0, 1, sum)
sort(degree, decreasing = TRUE)
```

hopeless	nothing_cheer	worthless	nervous
5	5	5	4
restless	everything_effort		
4	3		

```
# Node strength centrality
cent <- centrality(net)
sort(cent$strength, decreasing = TRUE)
```

NULL

```
centralityPlot(net)
```



1. Which symptom has highest degree?
2. Which has highest strength?
3. Do degree and strength tell a similar story?
4. How does this relate to edges?
5. Which symptoms might be leverage points to improve psychological distress?

3.9 Stability of centrality

```
# Case-dropping bootstrap for centrality stability
boot_cent <- bootnet(
  net,
  nBoots = 100,
  type = "case"
)
```

Note: bootnet will store only the following statistics: edge, strength, outStrength, inStrength

Bootstrapping...

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Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated

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Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated

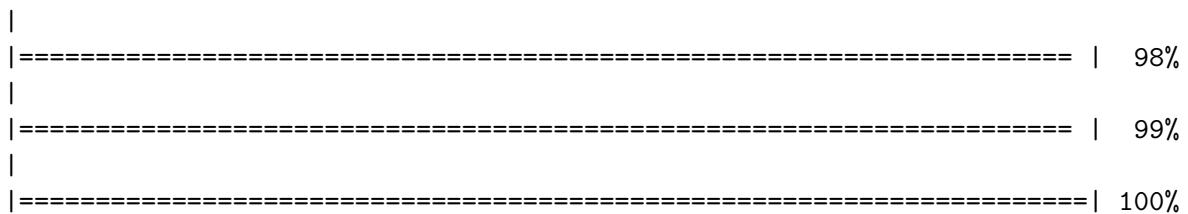
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Note: Network with lowest lambda selected as best network: assumption of sparsity might be violated



Computing statistics...



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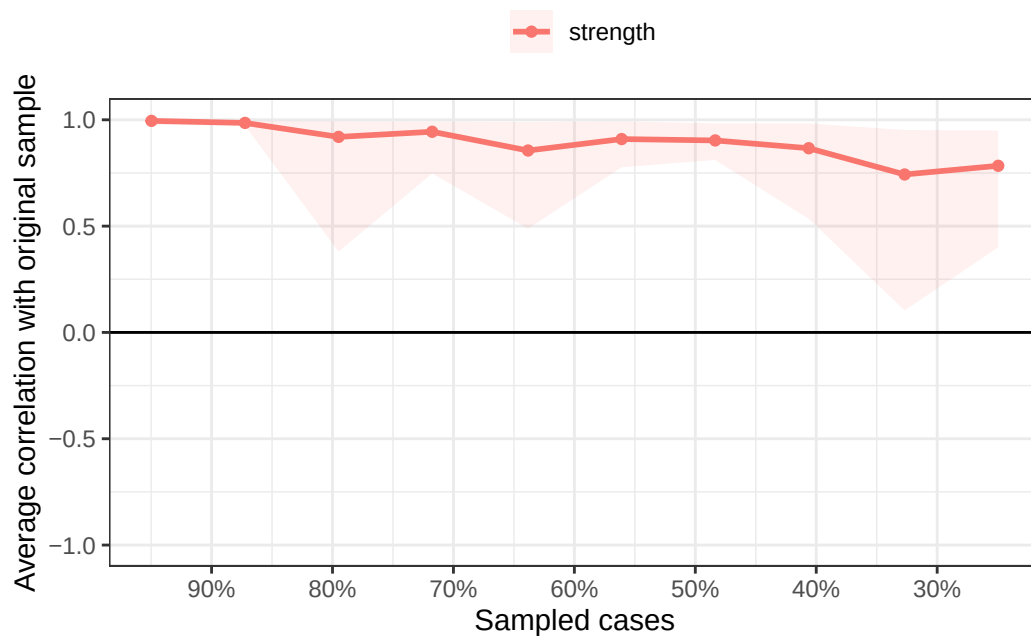
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```

# Plot centrality stability
plot(boot_cent, "strength")

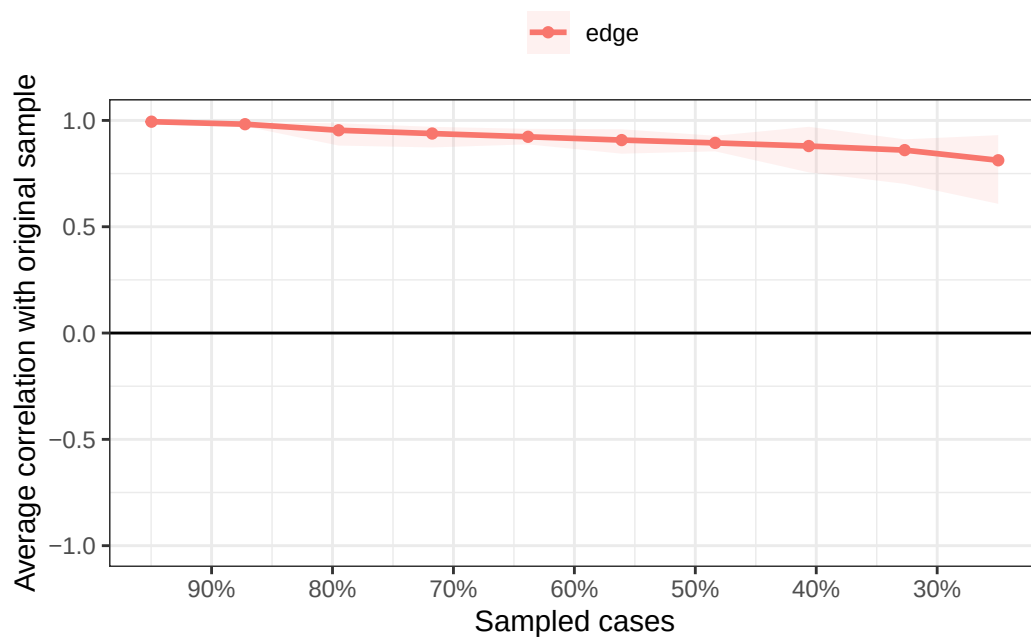
```



```

plot(boot_cent, "edge")

```



```
corStability(boot_cent)
```

=== Correlation Stability Analysis ===

Sampling levels tested:

	nPerson	Drop%	n
1	129	75.0	11
2	169	67.3	9
3	210	59.4	11
4	250	51.6	9
5	290	43.9	10
6	330	36.2	8
7	371	28.2	11
8	411	20.5	12
9	451	12.8	7
10	491	5.0	12

Maximum drop proportions to retain correlation of 0.7 in at least 95% of the samples:

edge: 0.673

- For more accuracy, run `bootnet(..., caseMin = 0.594, caseMax = 0.75)`

strength: 0.516

- For more accuracy, run `bootnet(..., caseMin = 0.439, caseMax = 0.594)`

Accuracy can also be increased by increasing both 'nBoots' and 'caseN'.

1. Does node strength remain consistent?
2. Do strongest edges stay stable?
3. Are edge weights and strength stable enough?
4. Is CS > .25?
5. What does this imply?

3.10 Network Comparison Test: High vs Low Income

We now test whether the psychological distress network during COVID-10 differs between those from low income households and those from low income households:

- **High-income** (`low_income = 0`)
- **Low-income** (`low_income = 1`)

The NCT compares:

- **Global strength**
- **Network structure**

- Edge differences

```
# Distress items - prepare the groups
distress <- c("nervous", "hopeless", "restless",
             "nothing_cheer", "everything_effort", "worthless")

# High-income group
items_high <- data[data$low_income == 0, distress]

# Low-income group
items_low <- data[data$low_income == 1, distress]

#clean the data
items_high_clean <- na.omit(items_high)
items_low_clean <- na.omit(items_low)

#run the NCT
nct <- NCT(
  data1 = items_high_clean,
  data2 = items_low_clean,
  test.edges = TRUE,
  test.centralities = FALSE,
  it = 1000
)
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```
summary(nct)
```

INDEPENDENT GROUPS GAUSSIAN NETWORK COMPARISON TEST

P-VALUE CORRECTION: none

NETWORK INVARIANCE TEST
Test statistic M: 0.2692764
p-value 0.1638362

GLOBAL STRENGTH INVARIANCE TEST
Global strength per group: 2.302893 2.233734
Test statistic S: 0.06915881

p-value 0.7512488

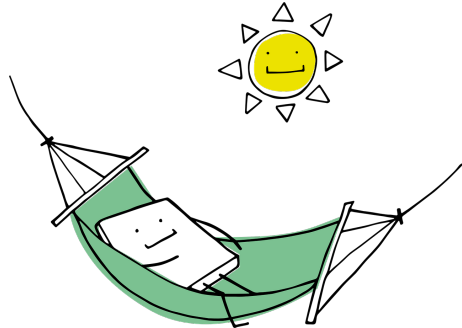
EDGE INVARIANCE TEST

	Var1	Var2	p-value	Test statistic	E
7	nervous	hopeless	0.20179820	0.15131636	
13	nervous	restless	0.40859141	0.10387309	
14	hopeless	restless	0.24875125	0.13803481	
19	nervous	nothing_cheer	0.31568432	0.07460678	
20	hopeless	nothing_cheer	0.87012987	0.02292007	
21	restless	nothing_cheer	0.22277722	0.12355624	
25	nervous	everything_effort	1.00000000	0.00000000	
26	hopeless	everything_effort	0.30469530	0.11986385	
27	restless	everything_effort	0.31168831	0.03184012	
28	nothing_cheer	everything_effort	0.01498501	0.26927637	
31	nervous	worthless	0.67032967	0.04931521	
32	hopeless	worthless	0.87412587	0.01870152	
33	restless	worthless	0.36863137	0.09146354	
34	nothing_cheer	worthless	0.09290709	0.20234389	
35	everything_effort	worthless	0.13786214	0.15688239	

1. Is there a significant difference in global strength?
2. Is there a difference in overall network structure?
3. Are any edges significantly different?
4. What does this suggest?

💡 Take a break!

If you haven't already taken a break today, we suggest you do so before you start the next section.



4 Part B: Apply Your Knowledge

4.1 Introduction to the dataset

You now need to repeat the same workflow using six wellbeing items from the Warwick–Edinburgh Mental Wellbeing Scale:

- optimistic
- useful
- deal_problems
- think_clearly
- feeling_close
- make_up_mind

5 Part C: Challenge Yourself — Combined Network

In the earlier sections, you estimated separate networks for six distress items and six wellbeing items. In this challenge, you will combine all 12 items into a single dataset and estimate a joint network model. This allows you to explore how distress and wellbeing processes interrelate within the same system.

You will then use `igraph` to identify clusters (communities) in the combined network.

5.1 Prepare data

```
# Vector of variable names
mental_health <- c("nervous", "hopeless", "restless",
                  "nothing_cheer", "everything_effort", "worthless",
                  "optimistic", "useful", "deal_problems", "think_clearly",
                  "feeling_close", "make_up_mind")

# Subset the actual data from your dataset
items <- data[mental_health]

str(items)
```

```
tibble [517 x 12] (S3: tbl_df/tbl/data.frame)
 $ nervous      : dbl+lbl [1:517] NA, 3, 2, 3, 3, 4, NA, NA, 5, NA, NA, 4, NA, 2...
 ..@ label      : chr "During the past 30 days, about how often did you feel... - nervous?"
 ..@ format.stata: chr "%40.0f"
 ..@ labels     : Named num [1:5] 1 2 3 4 5
 .. ..- attr(*, "names")= chr [1:5] "All of the time" "Most of the time" "Some of the time" "A
 $ hopeless     : dbl+lbl [1:517] NA, 3, 2, 3, 5, 4, NA, NA, 2, NA, NA, 4, NA, 2...
 ..@ label      : chr "During the past 30 days, about how often did you feel... - hopeless?"
 ..@ format.stata: chr "%40.0f"
 ..@ labels     : Named num [1:5] 1 2 3 4 5
 .. ..- attr(*, "names")= chr [1:5] "All of the time" "Most of the time" "Some of the time" "A
 $ restless     : dbl+lbl [1:517] NA, 4, 2, 3, 4, 5, NA, NA, 3, NA, NA, 5, NA, 2...
 ..@ label      : chr "During the past 30 days, about how often did you feel... - restless or
 ..@ format.stata: chr "%40.0f"
 ..@ labels     : Named num [1:5] 1 2 3 4 5
 .. ..- attr(*, "names")= chr [1:5] "All of the time" "Most of the time" "Some of the time" "A
 $ nothing_cheer : dbl+lbl [1:517] NA, 2, 5, 4, 5, 5, NA, NA, 1, NA, NA, 5, NA, 4...
```

```

..@ label      : chr "During the past 30 days, about how often did you feel... - so depressed"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "All of the time" "Most of the time" "Some of the time" "A
$ everything_effort: dbl+lbl [1:517] NA, 2, 4, 3, 4, 4, NA, NA, 1, NA, NA, 2, NA, 3...
..@ label      : chr "During the past 30 days, about how often did you feel... - that everyth
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "All of the time" "Most of the time" "Some of the time" "A
$ worthless     : dbl+lbl [1:517] NA, 4, 5, 5, 5, 5, NA, NA, 2, NA, NA, 3, NA, 3...
..@ label      : chr "During the past 30 days, about how often did you feel... - worthless?"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "All of the time" "Most of the time" "Some of the time" "A
$ optimistic    : dbl+lbl [1:517] NA, 3, 3, 2, 4, 4, NA, NA, 2, NA, NA, 3, NA, 3...
..@ label      : chr "I've been feeling optimistic about the future"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "None of the time" "Rarely" "Some of the time" "Often" ...
$ useful        : dbl+lbl [1:517] NA, 3, 3, 2, 4, 3, NA, NA, 1, NA, NA, 3, NA, 3...
..@ label      : chr "I've been feeling useful"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "None of the time" "Rarely" "Some of the time" "Often" ...
$ deal_problems : dbl+lbl [1:517] NA, 2, 4, 3, 4, 4, NA, NA, 2, NA, NA, 4, NA, 4...
..@ label      : chr "I've been dealing with problems well"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "None of the time" "Rarely" "Some of the time" "Often" ...
$ think_clearly : dbl+lbl [1:517] NA, 2, 5, 2, 4, 4, NA, NA, 1, NA, NA, 4, NA, 3...
..@ label      : chr "I've been thinking clearly"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "None of the time" "Rarely" "Some of the time" "Often" ...
$ feeling_close : dbl+lbl [1:517] NA, 3, 5, 1, 5, 4, NA, NA, 2, NA, NA, 4, NA, 3...
..@ label      : chr "I've been feeling close to other people"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "None of the time" "Rarely" "Some of the time" "Often" ...
$ make_up_mind  : dbl+lbl [1:517] NA, 3, 4, 3, 4, 4, NA, NA, 3, NA, NA, 3, NA, 4...
..@ label      : chr "I've been able to make up my mind about things"
..@ format.stata: chr "%40.0f"
..@ labels     : Named num [1:5] 1 2 3 4 5
.. ..- attr(*, "names")= chr [1:5] "None of the time" "Rarely" "Some of the time" "Often" ...

```

```
summary(items)
```

```
nervous      hopeless      restless      nothing_cheer
```


Min. :1.000	Min. :1.000	Min. :1.000	Min. :1.000
1st Qu.:2.000	1st Qu.:3.000	1st Qu.:2.000	1st Qu.:3.000
Median :3.000	Median :4.000	Median :3.000	Median :4.000
Mean :3.072	Mean :3.551	Mean :3.251	Mean :4.105
3rd Qu.:4.000	3rd Qu.:5.000	3rd Qu.:4.000	3rd Qu.:5.000
Max. :5.000	Max. :5.000	Max. :5.000	Max. :5.000
NA's :211	NA's :212	NA's :214	NA's :213
everything_effort	worthless	optimistic	useful
Min. :1.000	Min. :1.000	Min. :1.00	Min. :1.000
1st Qu.:3.000	1st Qu.:3.000	1st Qu.:2.00	1st Qu.:3.000
Median :3.000	Median :4.000	Median :3.00	Median :3.000
Mean :3.326	Mean :4.057	Mean :2.88	Mean :3.156
3rd Qu.:4.000	3rd Qu.:5.000	3rd Qu.:3.00	3rd Qu.:4.000
Max. :5.000	Max. :5.000	Max. :5.00	Max. :5.000
NA's :216	NA's :221	NA's :208	NA's :209
deal_problems	think_clearly	feeling_close	make_up_mind
Min. :1.00	Min. :1.00	Min. :1.000	Min. :1.00
1st Qu.:3.00	1st Qu.:3.00	1st Qu.:2.000	1st Qu.:3.00
Median :3.00	Median :3.00	Median :3.000	Median :4.00
Mean :3.33	Mean :3.33	Mean :3.104	Mean :3.54
3rd Qu.:4.00	3rd Qu.:4.00	3rd Qu.:4.000	3rd Qu.:4.00
Max. :5.00	Max. :5.00	Max. :5.000	Max. :5.00
NA's :208	NA's :208	NA's :208	NA's :208

```
# Ensure numeric type
items <- data.frame(lapply(items, function(x) as.numeric(as.character(x))))
```

5.2 Estimate the 12-item network and Plot it

```
net <- estimateNetwork(items,
  default = "EBICglasso",
  corMethod = "cor_auto"
)
```

Estimating Network. Using package::function:

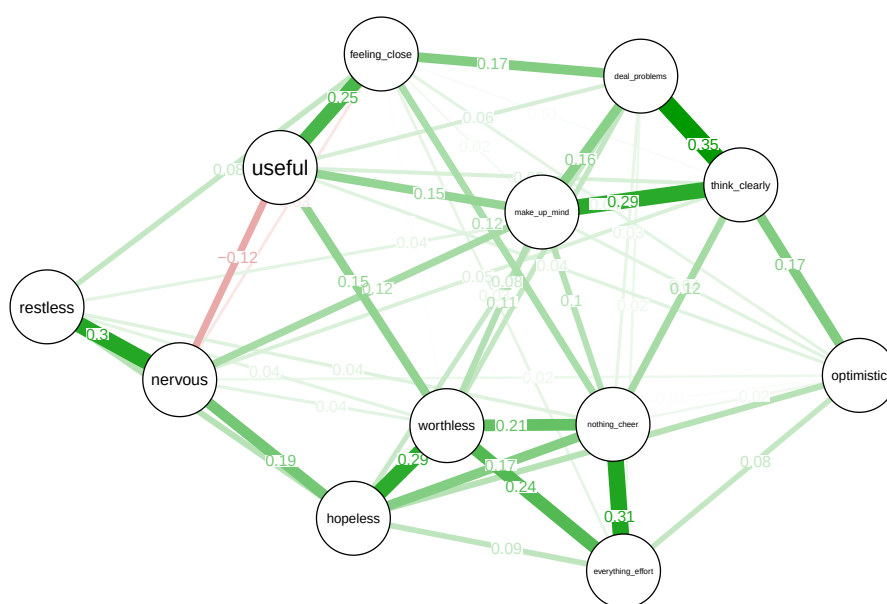
- qgraph::EBICglasso for EBIC model selection
- using glasso::glasso
- qgraph::cor_auto for correlation computation
- using lavaan::lavCor

Variables detected as ordinal: nervous; hopeless; restless; nothing_cheer; everything_effort; wor

Warning in EBICglassoCore(S = S, n = n, gamma = gamma, penalize.diagonal =
penalize.diagonal, : A dense regularized network was selected (lambda < 0.1 *

lambda.max). Recent work indicates a possible drop in specificity. Interpret the presence of the smallest edges with care. Setting threshold = TRUE will enforce higher specificity, at the cost of sensitivity.

```
qgraph(
  net$graph,
  labels = colnames(items),
  layout = "spring",
  edge.labels = TRUE,          # <-- this adds the numbers
  edge.label.cex = 0.8        # <-- adjust text size (optional)
)
```



5.3 Convert to igraph and detect clusters using various different algorithms

```
W <- net$graph # <-- convert to absolute values to get rid of negative values
W_abs <- abs(W) # <-- convert to absolute values to get rid of negative values

g <- graph_from_adjacency_matrix(W_abs, mode = "undirected", weighted = TRUE, diag = FALSE)

wc <- cluster_walktrap(g, weights = E(g)$weight)
lv <- cluster_louvain(g, weights = E(g)$weight)
fg <- cluster_fast_greedy(g, weights = E(g)$weight)
im <- cluster_infomap(g, e.weights = E(g)$weight)
le <- cluster_leading_eigen(g, weights = E(g)$weight)
```

5.4 Inspect Clusters

membership(wc)

nervous	hopeless	restless	nothing_cheer
3	2	3	2
everything_effort	worthless	optimistic	useful
2	2	1	1
deal_problems	think_clearly	feeling_close	make_up_mind
1	1	1	1

membership(lv)

nervous	hopeless	restless	nothing_cheer
1	2	1	2
everything_effort	worthless	optimistic	useful
2	2	3	1
deal_problems	think_clearly	feeling_close	make_up_mind
3	3	1	3

membership(fg)

nervous	hopeless	restless	nothing_cheer
1	3	1	3
everything_effort	worthless	optimistic	useful
3	3	2	1
deal_problems	think_clearly	feeling_close	make_up_mind
2	2	1	2

membership(im)

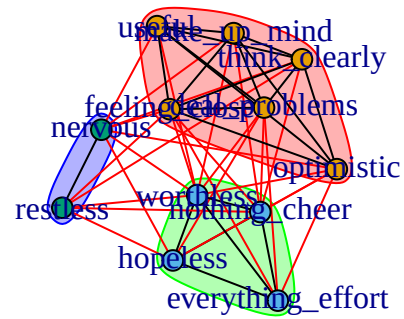
nervous	hopeless	restless	nothing_cheer
1	1	1	1
everything_effort	worthless	optimistic	useful
1	1	1	1
deal_problems	think_clearly	feeling_close	make_up_mind
1	1	1	1

membership(le)

nervous	hopeless	restless	nothing_cheer
1	1	1	3
everything_effort	worthless	optimistic	useful
3	3	2	2
deal_problems	think_clearly	feeling_close	make_up_mind
2	2	2	2

5.5 Plot each solution

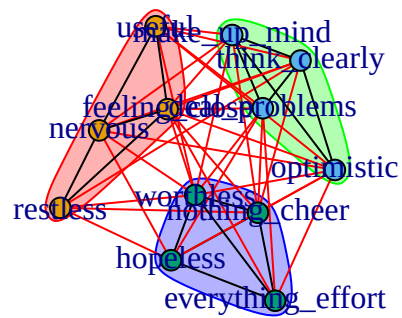
```
plot(wc, g)
```



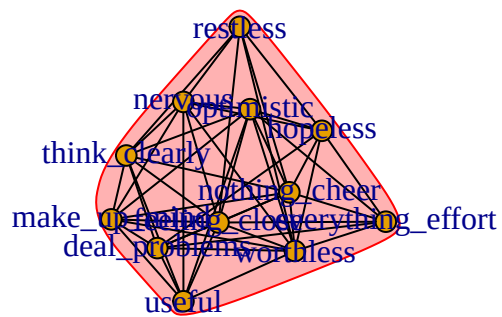
```
plot(lv, g)
```



```
plot(fg, g)
```



```
plot(im, g)
```



5.6 Compare the number of clusters

```
length(unique(membership(wc)))
```

```
[1] 3
```

```
length(unique(membership(lv)))
```

```
[1] 3
```

```
length(unique(membership(fg)))
```

```
[1] 3
```

```
length(unique(membership(im)))
```

```
[1] 1
```

```
length(unique(membership(le)))
```

```
[1] 3
```

1. How many clusters does igraph detect?
2. Do distress and wellbeing items mix together or separate cleanly?
3. What does this tell you about the structure of mental health?

6 End of practical

! Feedback!

These practical sessions are new this year. Please take a moment to rate the difficulty and length of these practical activities via [this survey](#), and let us know of any issues you encountered or aspects you didn't understand (positive feedback is also welcome!). You will need to be logged in with your university account to respond, but your submission will be anonymous.

For more general questions about the practical, please post them on the [VLE Discussion Board](#).